# Federated Learning on Heterogeneous Sensor Data

By David, Jonathan Truong Son

# Federated Learning Implementation

Libraries/Methods

- Pytorch, CNN, 15 clients, Stochastic Gradient Descent (SGD)

Datasets

- MNIST: handwritten digits(0-9), 60k, 28x28 pixels, grayscale

Levels of Heterogeneity

- Determined by the bias variable (0-1)
- Affects the data distribution to clients

```
# assign a data point to a group
upper_bound = (y) * (1 - bias_weight) / 9. + bias_weight
lower_bound = (y) * (1 - bias_weight) / 9.
rd = np.random.random_sample()
```

# Level of Heterogeneity (Bias)

Fang Distribution Characteristics:

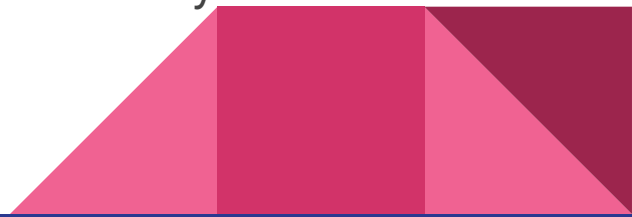
- Skews data distribution among clients based on the bias_weight parameter.

Logic:

- If rd > upper_bound: Assign the data point to a group after y.
- If rd < lower_bound: Assign the data point to a group before y.
- Otherwise: Assign the data point directly to group y.

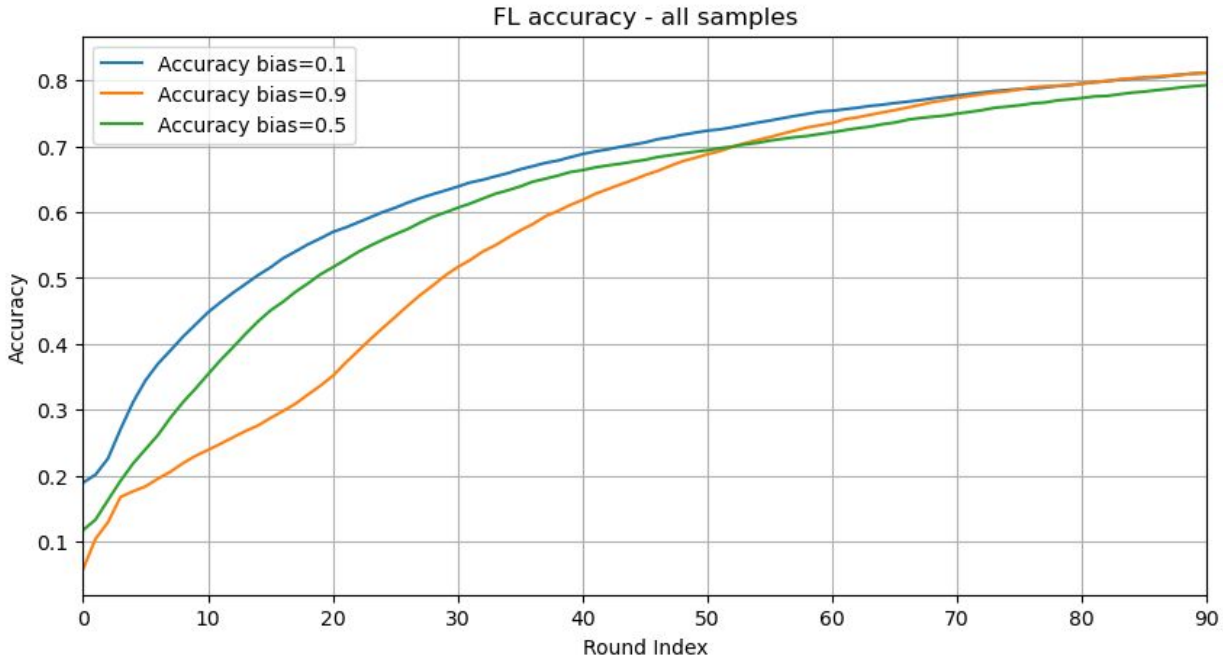
Bias_weight = k (0<k<1):

- (k*100)% of the data client decision is influenced by the label y.
- 1 - (k*100)% is determined by randomness.



# Accuracy at different Data distribution

| Bias | Group tendency | Data Distribution Type |
|---|---|---|
| 0.1 | Mostly random | Close to IID |
| 0.5 | Balanced between fixed label and randomness | half-IID |
| 0.9 | Strong label biased | Non-IID |



# Attacks

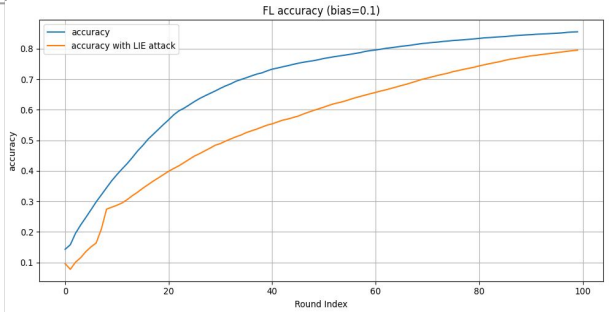
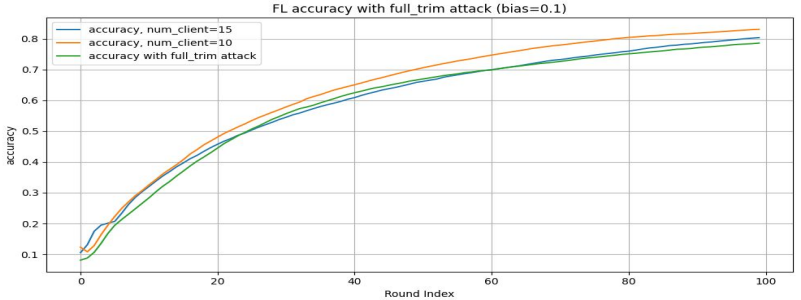
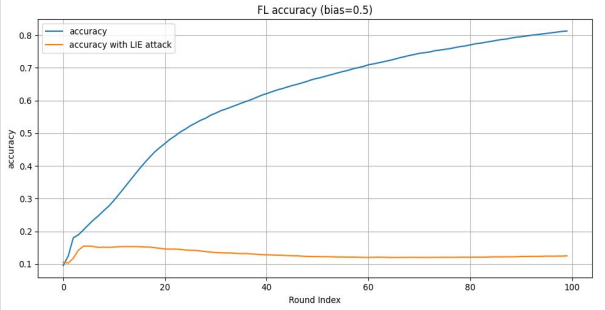
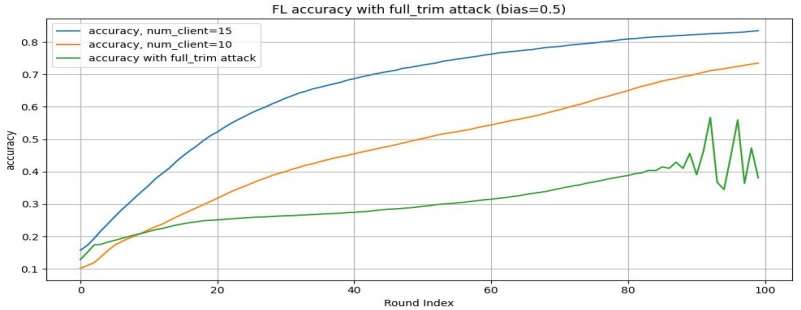
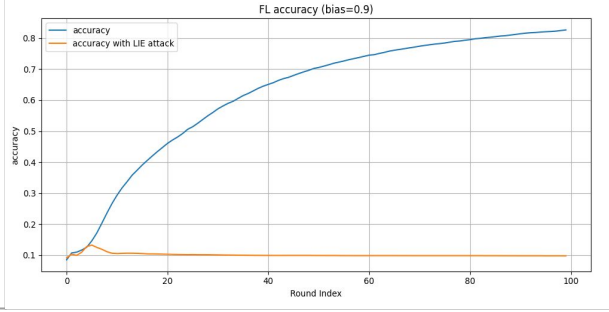
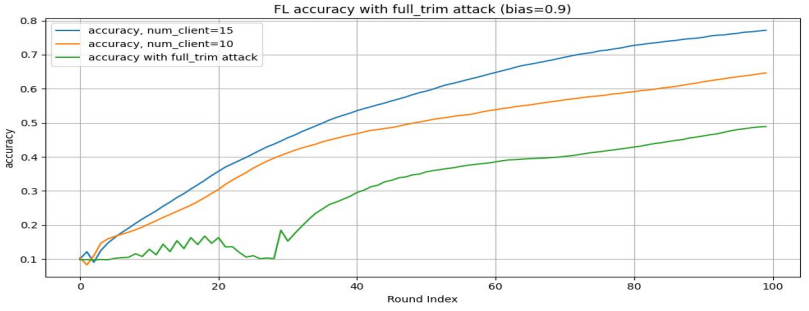
- Full knowledge Static AGR Attack
  - The attacker has full knowledge of the gradient from all client
  - Depend on the number of compromised clients (number of client be attacked)
  - the adversary manipulates the model updates in such a way that it can effectively change the global model after aggregation, without being easily detected

- LIE Attack
  - The LIE attack modifies the gradient aggregation by using the average and standard deviation of the collected gradients.
  - The attack's strength is controlled by the parameter z. Larger values of z introduce more perturbation into the gradient.

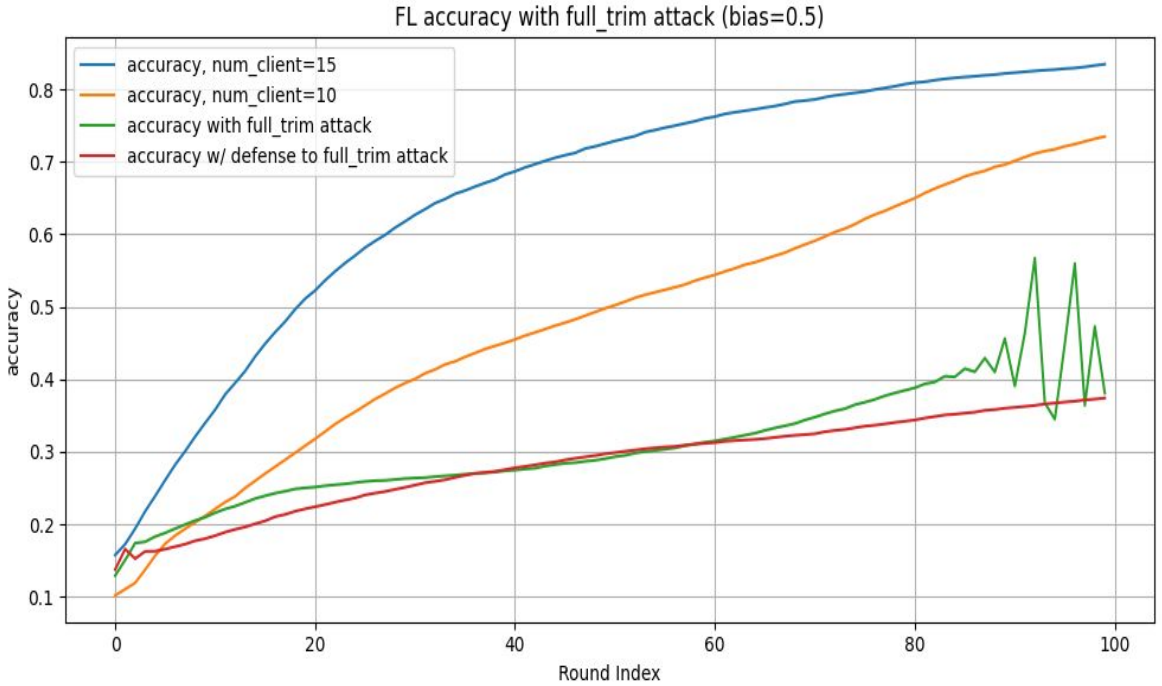
# Attack codes

```python
def full_trim(v, f):
    vi_shape = v[0].unsqueeze(0).T.shape
    v_tran = v.T

    maximum_dim = torch.max(v_tran, dim=1)
    maximum_dim = maximum_dim[0].reshape(vi_shape)
    minimum_dim = torch.min(v_tran, dim=1)
    minimum_dim = minimum_dim[0].reshape(vi_shape)
    direction = torch.sign(torch.sum(v_tran, dim=-1, keepdims=True))
    directed_dim = (direction > 0) * minimum_dim + (direction < 0) * maximum_dim

    for i in range(f):
        # apply attack to compromised worker devices with randomness
        random_12 = 2
        tmp = directed_dim * ((direction * directed_dim > 0) / random_12 + (direction * directed_dim < 0) * random_12)
        tmp = tmp.squeeze()
        v[i] = tmp
    return v
```

```python
#LIE Attack
def lie_attack(all_updates, z):
    avg = torch.mean(all_updates, dim=0)
    std = torch.std(all_updates, dim=0)
    return avg + z * std
```

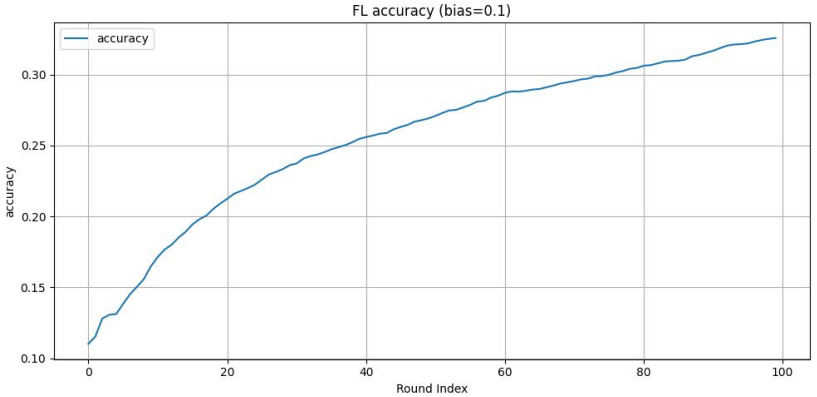
| Bias | Comparison (all, LIE attack) | Comparison (all, 10/15, FKTrim attack) |
| --- | --- | --- |
| 0.1 |  |  |
| 0.5 |  |  |
| 0.9 |  |  |

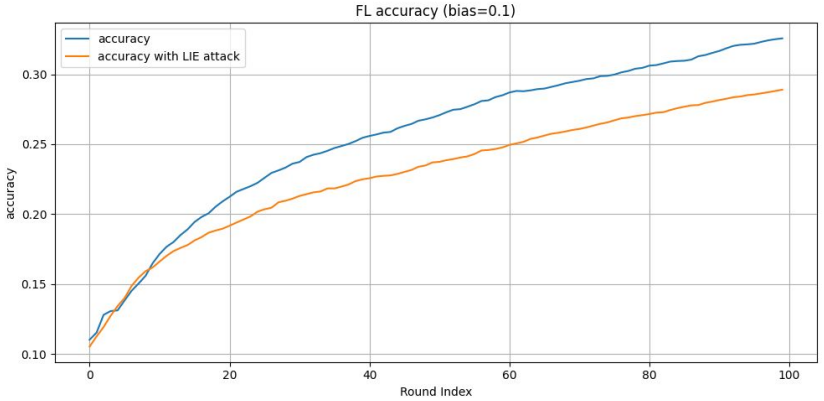
# tr_mean for prevent full_trim attack (bias = 0.5)

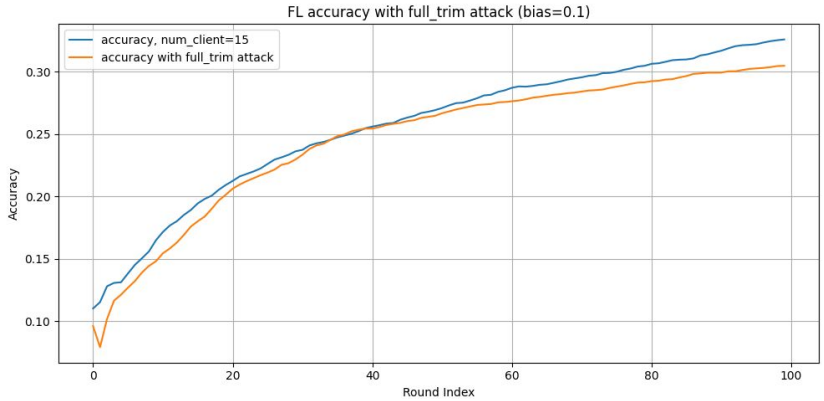


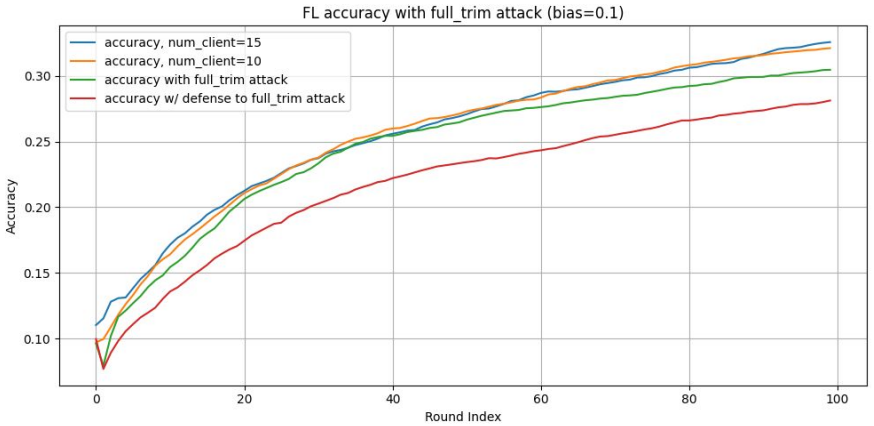
# Thank You.

# CIFAR10, bias=0.1

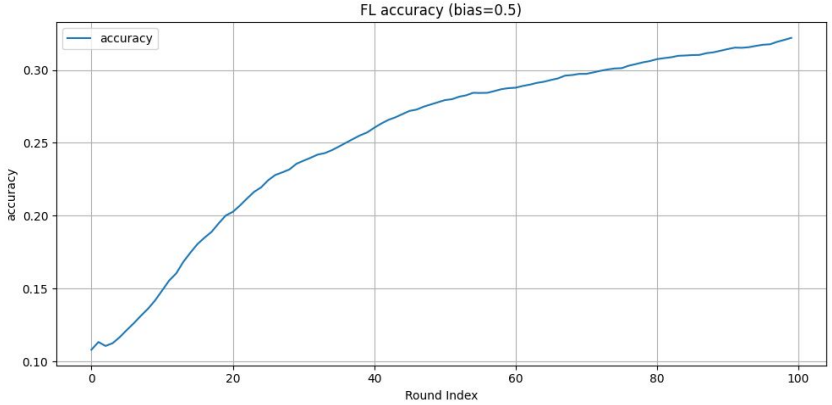


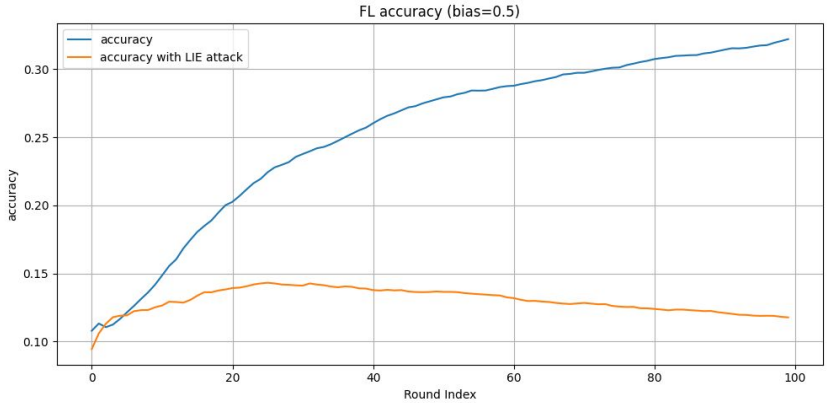



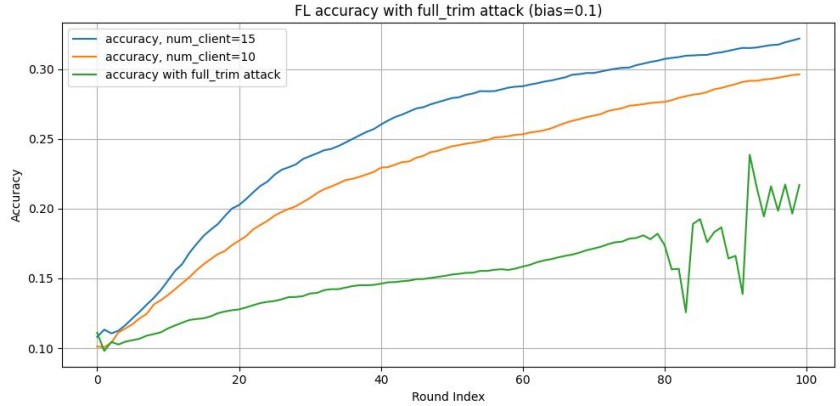



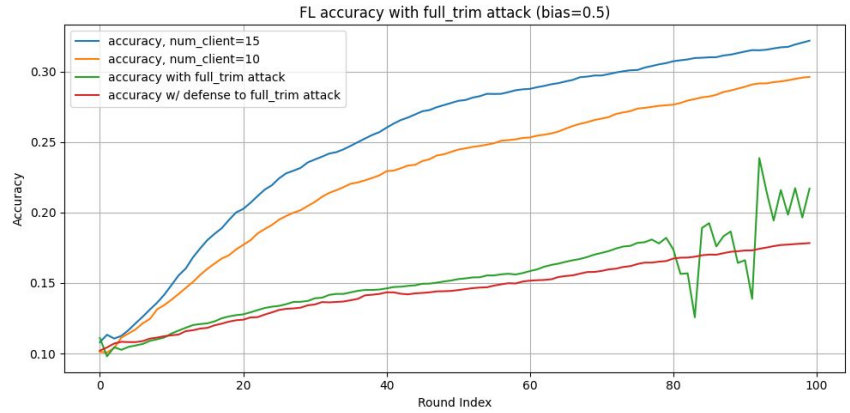



# CIFAR10, bias=0.5

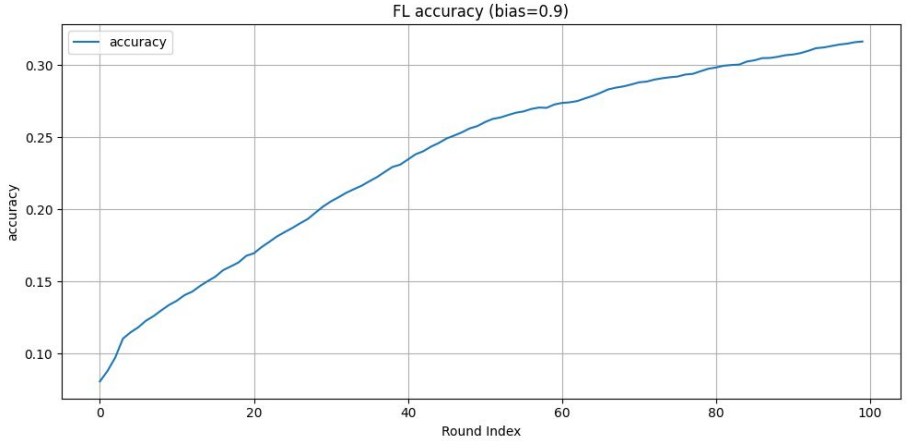


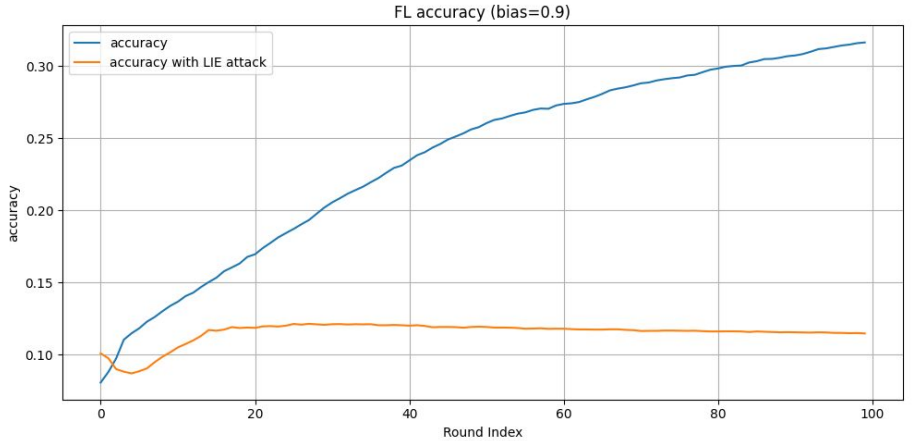



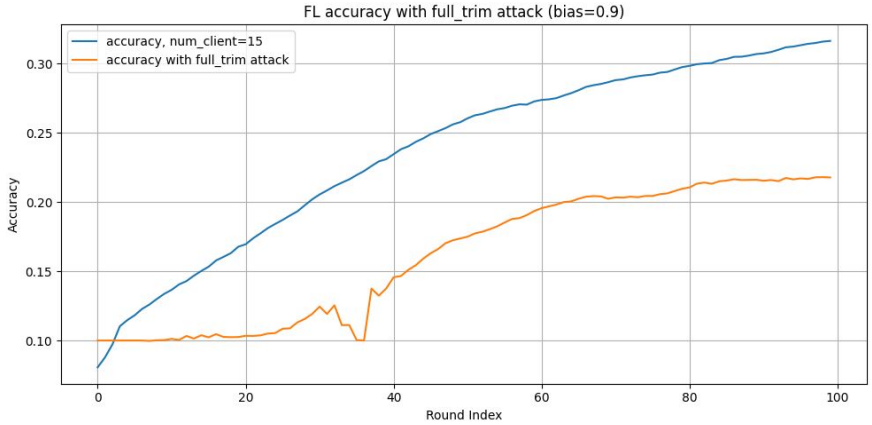



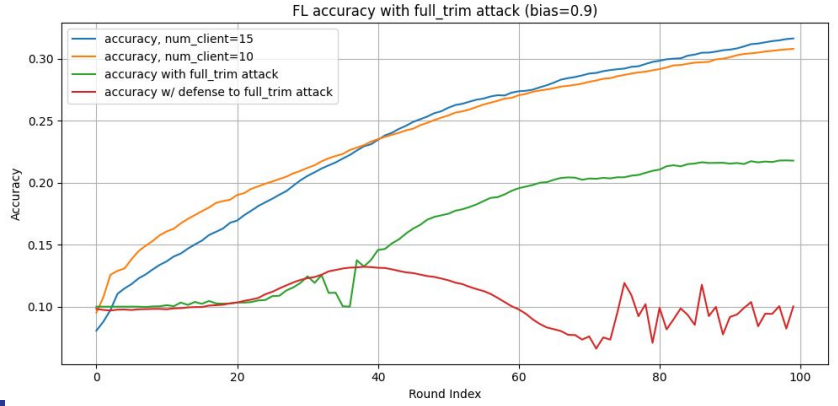



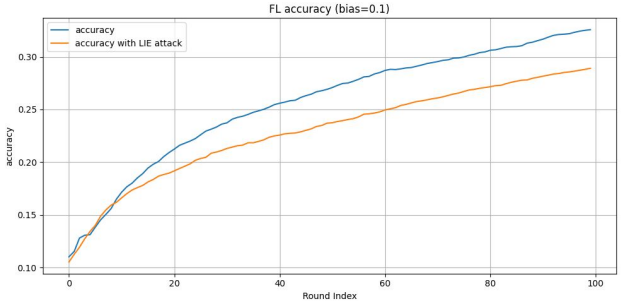
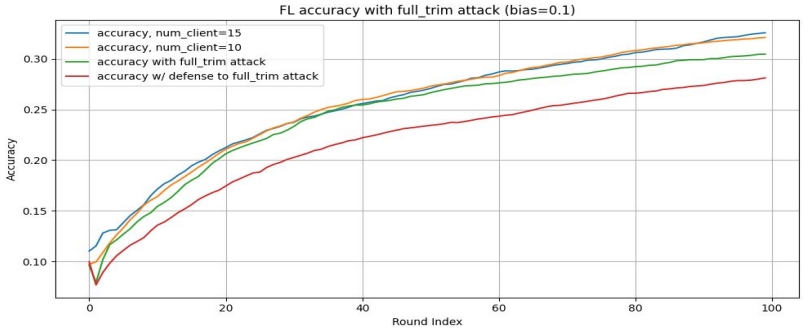
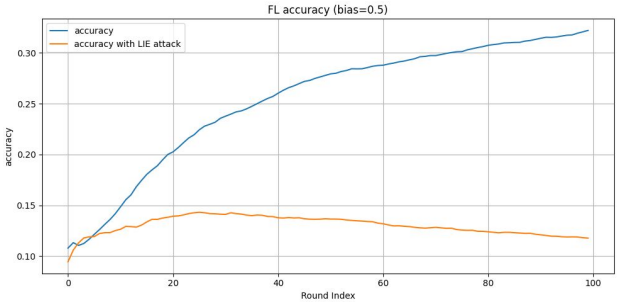
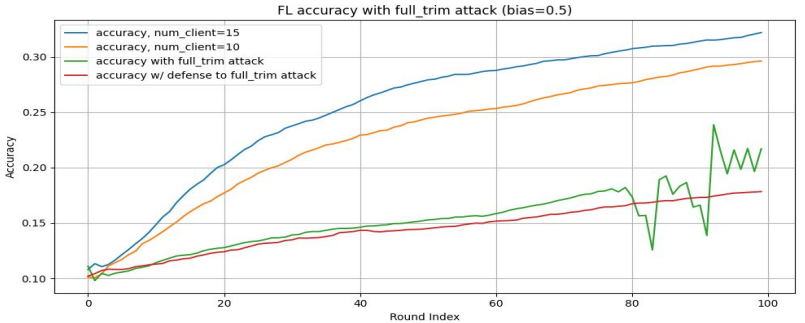
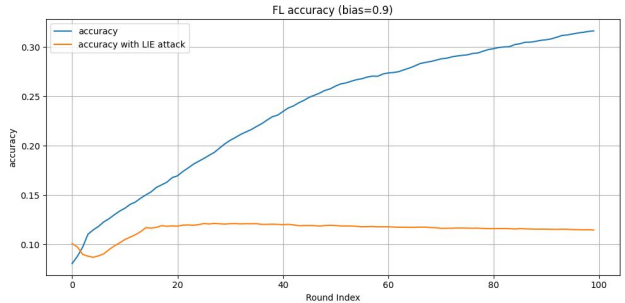
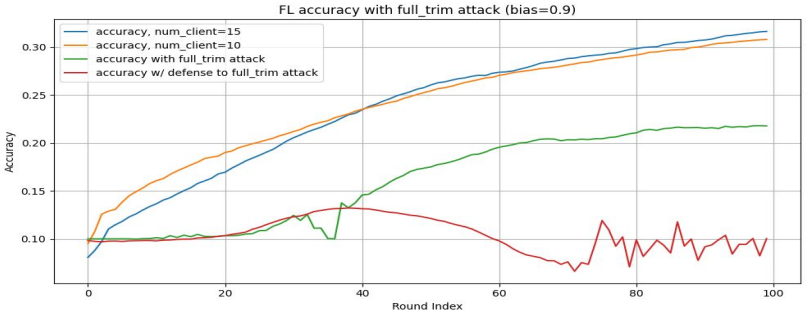
# CIFAR10, bias=0.9



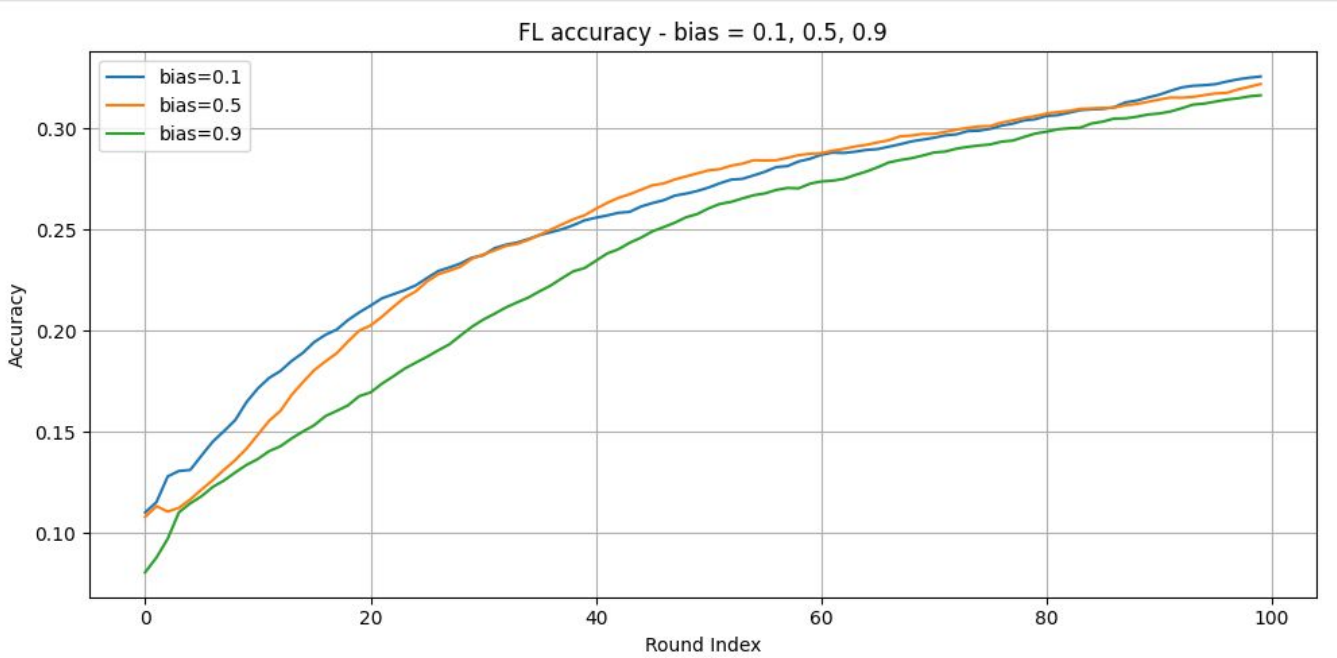



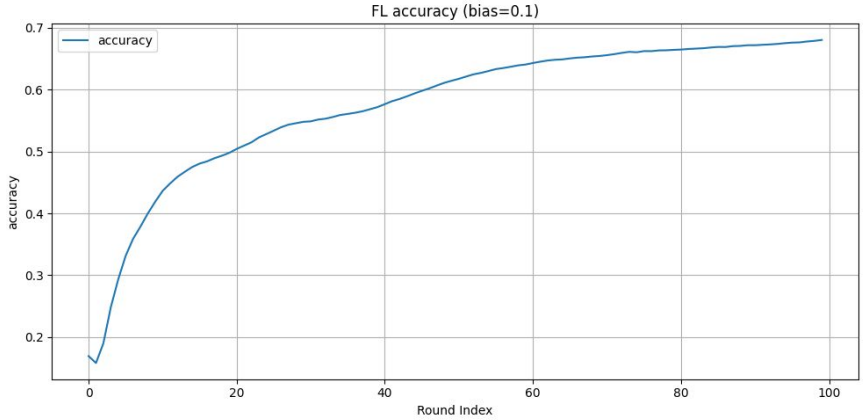



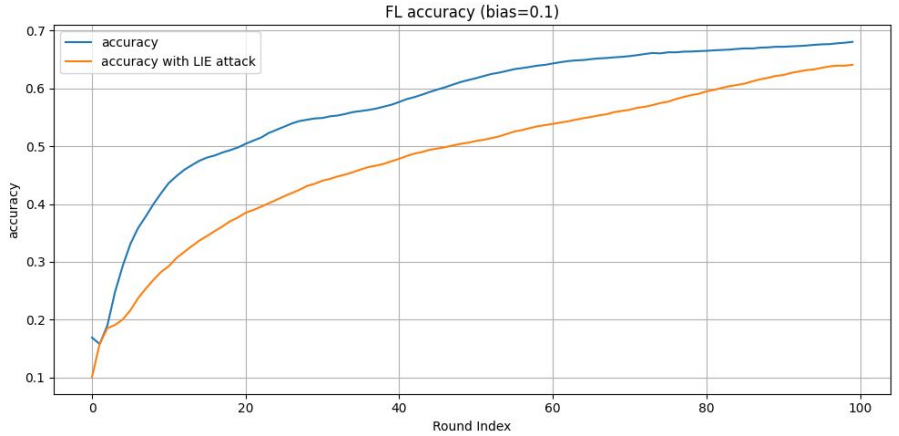
| Bias | Comparison (all, LIE attack) | Comparison (all, 10/15, FKTrim attack) |
|---|---|---|
| 0.1 |  |  |
| 0.5 |  |  |
| 0.9 |  |  |

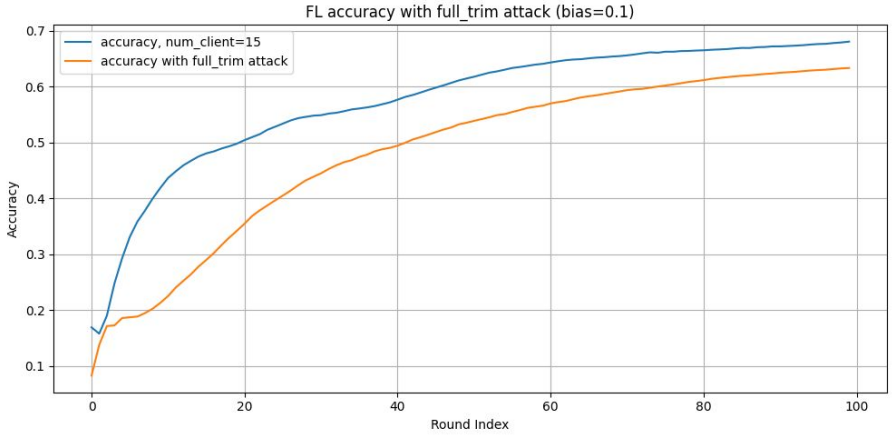
# CIFAR10, bias = 0.1, 0.5, 0.9

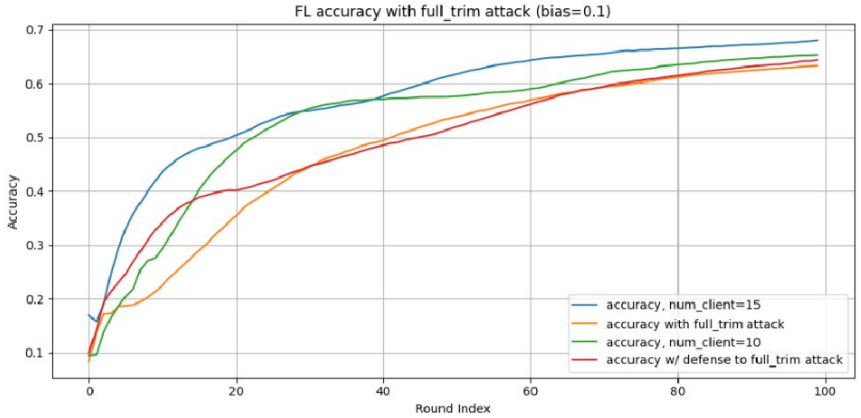


# Fashion MNIST, bias = 0.1



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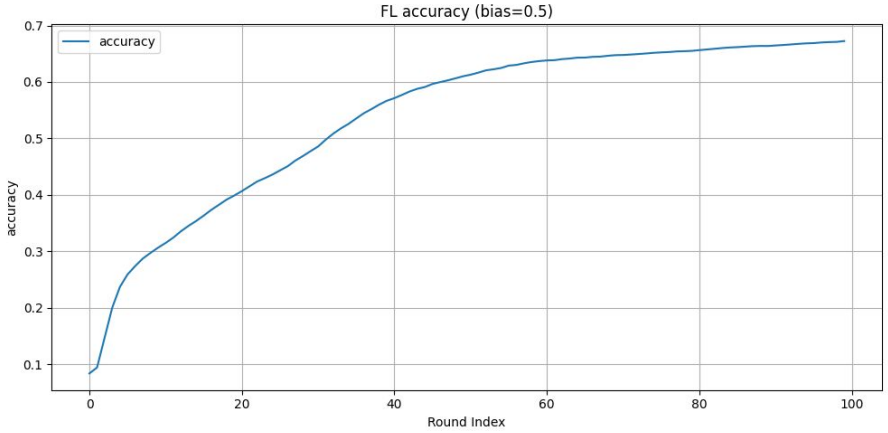


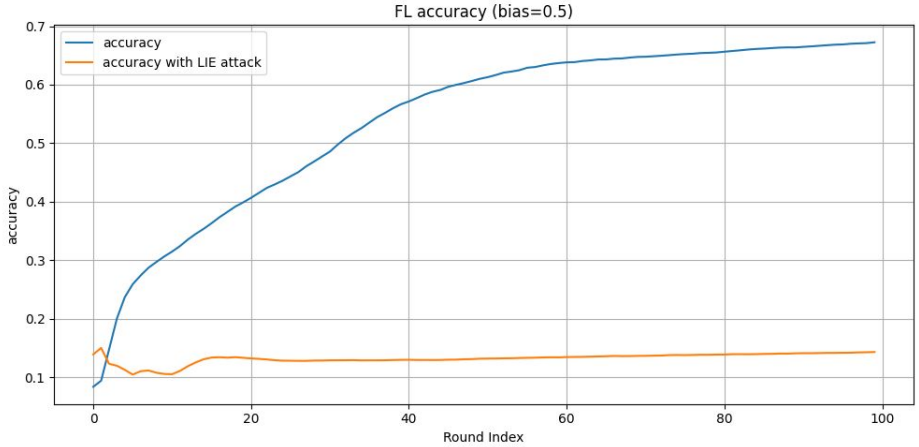


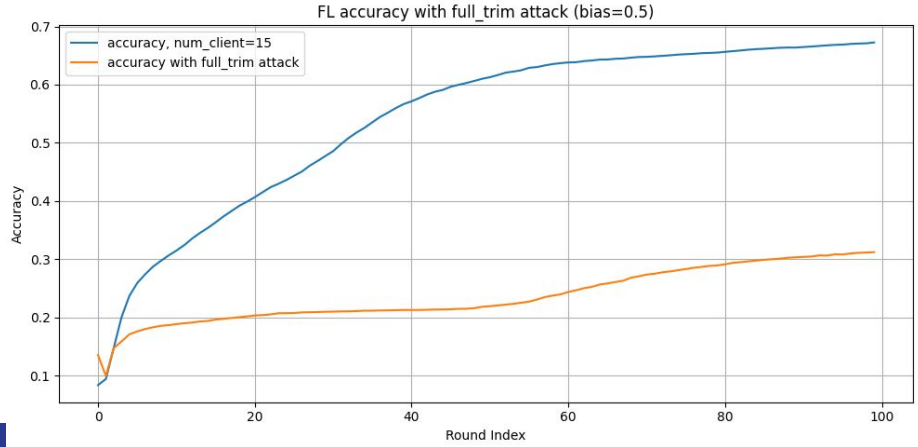


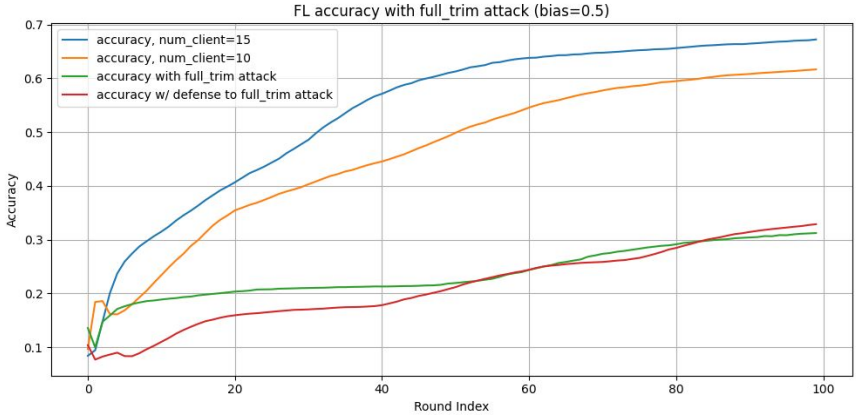

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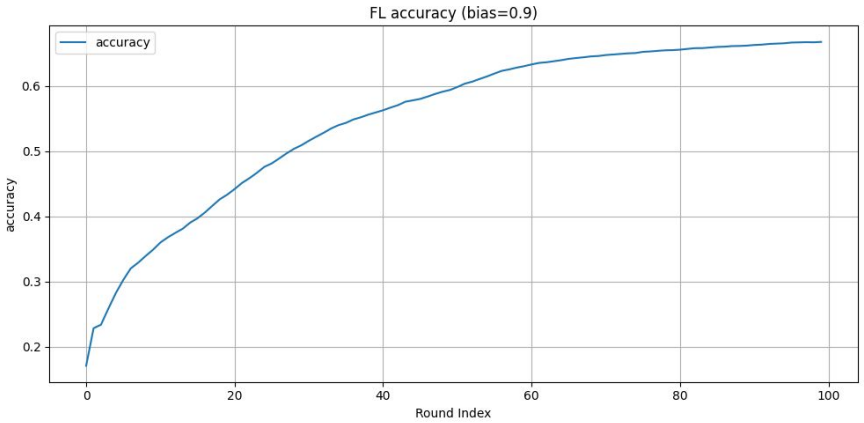


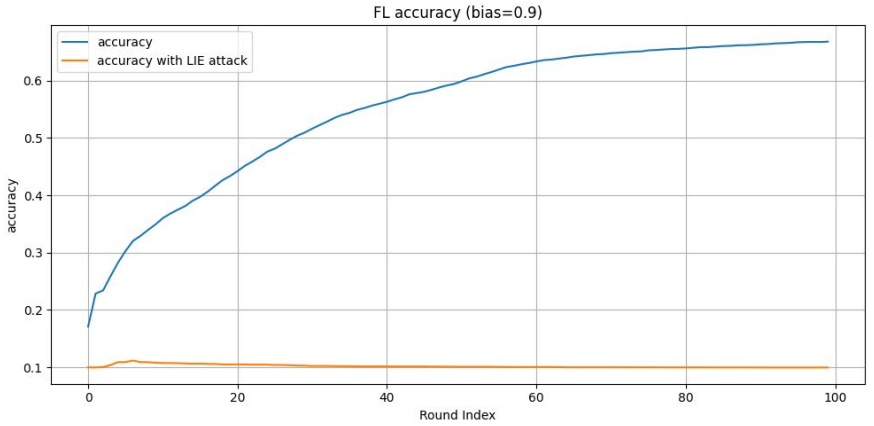


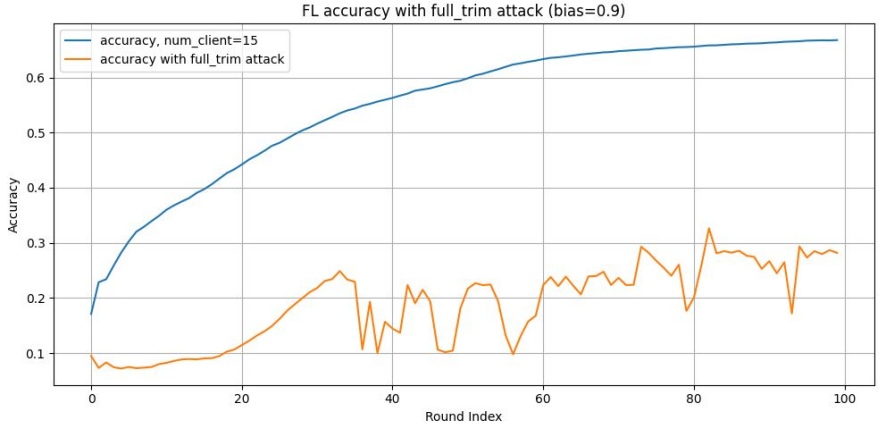


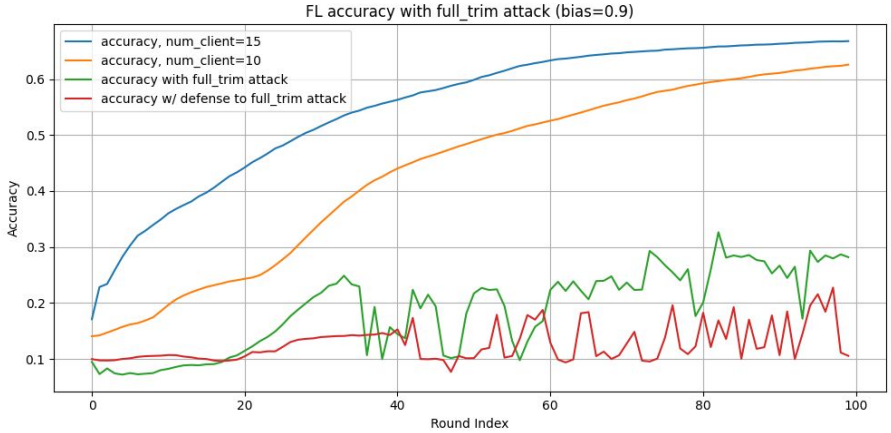

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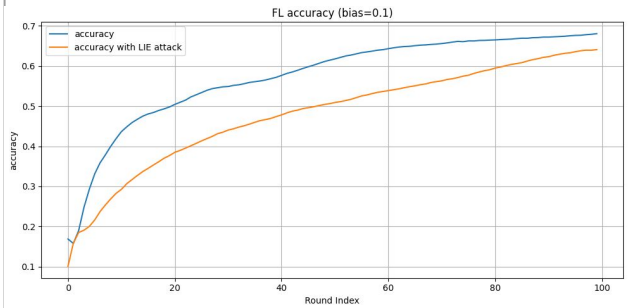
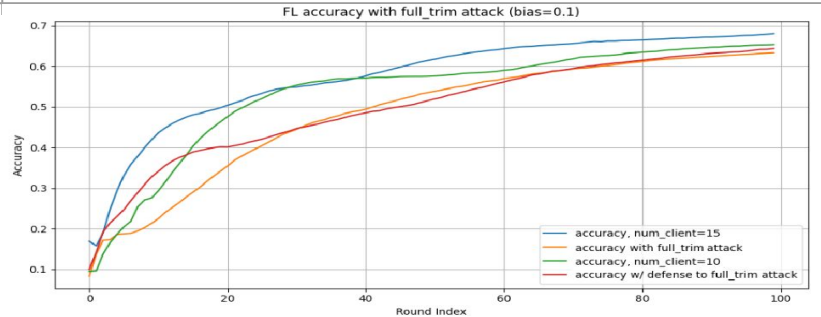
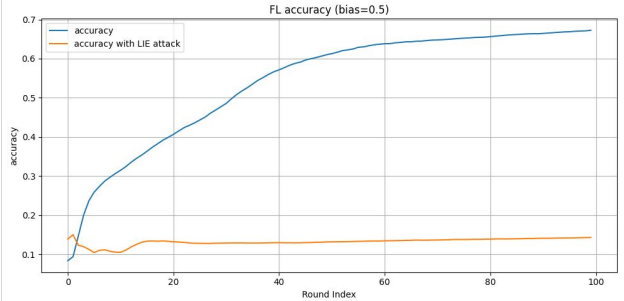
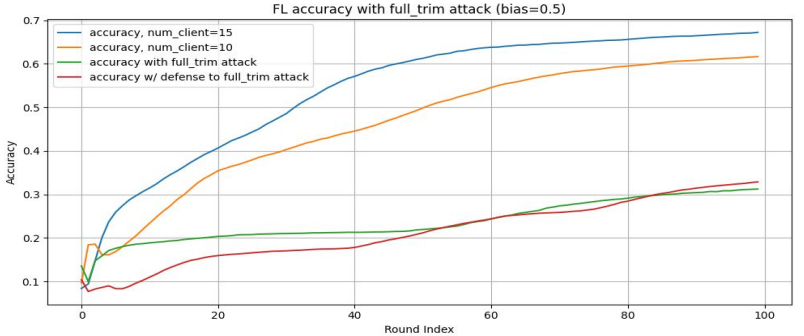
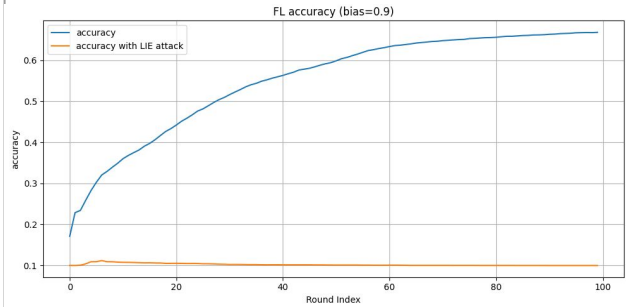
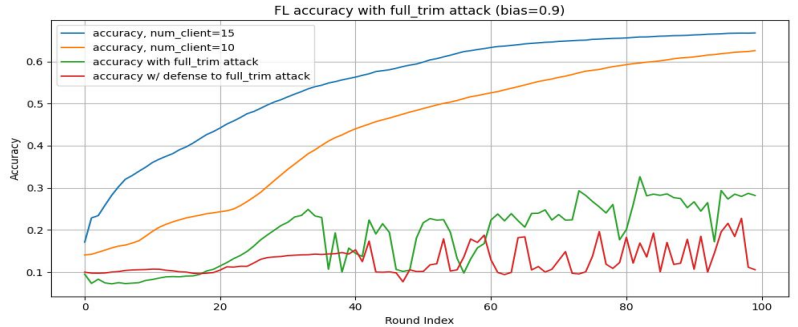
| 0.1 |  |  |
| 0.5 |  |  |
| 0.9 |  |  |

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